30 Days SQL Micro course answer pdf

1. Total Number of Patients

Find the total number of patients across all hospitals.

Answer:-

```
SELECT SUM(patients_count) AS total_patients
FROM hospital;
```

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2. Average Number of Doctors per Hospital

Retrieve the average count of doctors available in each hospital.

Answer:-

```
SELECT hospital_name, AVG(doctors_count) AS avg_doctors
FROM hospital
GROUP BY hospital_name;
```

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3. Top 3 Departments with the Highest Number of Patients

Find the top 3 hospital departments that have the highest number of patients.

Answer:-

```
SELECT department, SUM(patients_count) AS total_patients
FROM hospital
GROUP BY department
ORDER BY total_patients DESC
LIMIT 3;
```

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4. Hospital with the Maximum Medical Expenses

Identify the hospital that recorded the highest medical expenses.

Answer:-

```
SELECT hospital_name, SUM(medical_expenses) AS
total_expense
FROM hospital
GROUP BY hospital_name
ORDER BY total_expense DESC
LIMIT 1;
```

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5. Daily Average Medical Expenses

Calculate the average medical expenses per day for each hospital.

Answer:-

```
SELECT hospital_name,
ROUND(AVG(medical_expenses / NULLIF(discharge_date -
admission_date, 0)),2) AS avg_expense
FROM hospital
GROUP BY hospital_name;
```

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6. Longest Hospital Stay

Find the patient with the longest stay by calculating the difference between Discharge Date and Admission Date.

Answer:-

```
SELECT hospital_name,
discharge_date - admission_date AS longest_stay
FROM hospital
ORDER BY longest_stay DESC
LIMIT 1;
```

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7. Total Patients Treated Per City

Count the total number of patients treated in each city.

Answer:-

```
SELECT location, SUM(patients_count) AS total_patients
FROM hospital
GROUP BY location;
```

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8. Average Length of Stay Per Department

Calculate the average number of days patients spend in each department.

Answer:-

```
SELECT department,
ROUND(AVG(discharge_date - admission_date),2) AS
avg_stay_days
FROM hospital
GROUP BY department;
```

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9. Department with the Lowest Number of Patients

Find the department with the least number of patients.

Answer:-

```
SELECT department, SUM(patients_count) AS total_patients
FROM hospital
GROUP BY department
ORDER BY total_patients ASC
LIMIT 1;
```

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10. Monthly Medical Expenses Report

Group the data by month and calculate the total medical expenses for each month.

Answer:-

```
SELECT EXTRACT(MONTH FROM admission_date) AS month,
SUM(medical_expenses) AS monthly_expense
FROM hospital
GROUP BY month
ORDER BY month;
```